Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Next-Generation Electrophysiology Platform for Human Organoid-Based Functional Assessment			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio · Medical		Electrical · Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices	Measuring Instruments	Measurement Sensors and Components
Concept of Proposed Technology	The proposed technology is a next-generation, electrophysiology platform designed to enable scalable, quantitative, standardized, and reproducible functional assessment using both two-dimensional (2D) cell cultures and three-dimensional (3D) human organoid models. Global regulatory and scientific trends increasingly emphasize human-relevant, non-animal–based evaluation methodologies, including New Approach Methodologies (NAMs) that integrate advanced *in vitro* human models with computational analysis[1]. Despite this shift, existing microelectrode array (MEA) systems remain limited in scalability, throughput, and analytical depth, particularly for complex 3D organoid models[2]. Current platforms largely focus on signal acquisition with minimal automated interpretation, resulting in limited reproducibility and difficulty translating electrophysiological data into decision-ready outputs. This project addresses these limitations by integrating electrophysiological measurement, GPU-accelerated signal processing, and embedded automated electrophysiology analytics pipeline into a			

	unified platform. The technology is designed to automatically transform raw time-series electrophysiological data into standardized functional metrics, enabling cross-condition comparison, dose–response analysis, and non-clinical decision support. Key technological elements include: • Support for high-density, multi-well 2D and 3D MEA formats suitable for high-throughput workflows • Scalable, high-performance signal acquisition and processing architectures • Embedded signal processing and electrophysiology analytics pipeline for automated interpretation of rhythm, conduction, and network-level activity The platform is intended for applications such as organoid-based functional pharmacology, cardiotoxicity and neurotoxicity assessment, and broader drug discovery and development research, while also serving as a versatile electrophysiology research system aligned with global regulatory expectations.
Rationale for Joint R&D Project	Development of a next-generation, high-throughput electrophysiology platform requires the integration of multiple specialized technical domains, including bioengineering, electrophysiology, signal processing, and computational analytics. In particular, 3D MEA systems compatible with human cardiac and neuronal organoids remain technically challenging and are available only at a limited number of overseas research institutions with established organoid culture and electrophysiology expertise. Reproducing this level of capability independently within the domestic industrial ecosystem would require significant time and investment, increasing technical uncertainty and delaying feasibility validation. International joint R&D enables efficient access to research-grade 3D MEA concepts, organoid-based proof-of-concept validation, and scientific risk reduction, while domestic SMEs contribute complementary strengths in system engineering, product-oriented MEA hardware development, embedded analytics implementation, and commercialization. This clear division of roles reduces duplicated effort, accelerates development, and supports rapid transition from research-grade feasibility to commercial-grade electrophysiology systems. The project aligns with national and global priorities by advancing human-relevant testing technologies, strengthening domestic electrophysiology instrumentation capabilities, and enhancing Korea's competitiveness in next-generation life science tools.
Key Words	1. 3D Microelectrode Array (MEA) 2. High-Throughput Electrophysiology Platform 3. GPU-Accelerated Signal Processing 4. 3D Electrophysiological Conduction Mapping 5. Human Organoid-Based Functional Assessment

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)

R&D Objectives

Overall Objective

To develop a commercially deployable, electrophysiology platform integrating advanced 2D and 3D MEA devices, a high-throughput reader system, and GPU-accelerated signal processing and data analysis, and to advance the technology from TRL 3 to TRL 6.

Specific Objectives

- Develop advanced multi-well 3D MEA plate technologies suitable for stable electrophysiological recording from human organoid models
- Develop an ultra-high-throughput 2D MEA plate platform (e.g., 384-well format) for scalable electrophysiological screening
- Design and implement a high-channel-count, high-sampling-rate MEA reader system supporting both 2D and 3D formats
- Develop an electrophysiology analytics engine for automated signal preprocessing, feature extraction, and quantitative interpretation
- Integrate GPU-accelerated parallel signal processing to support high-throughput data analysis
- Validate platform performance using human cardiac and neuronal organoid–based functional pharmacology use cases

R&D Target Specifications

Category	Performance Metric	Target Specification
1. 3D MEA Multiwell Plate	Electrode density per well	**≥ 8 electrodes per well** for stable acquisition of 3D electrophysiological signals from human organoid models
2. 2D MEA Multiwell Plate	Cross-well reproducibility	**Coefficient of variation(CV) ≤ 20%** for key electrophysiological parameters (e.g., firing rate, beat rate) across 384 wells under identical experimental conditions
3. MEA Multiwell Reader	Sampling rate	**≥ 10 kHz per channel**, suitable for high-fidelity cardiac and neuronal electrophysiology measurements
4. MEA Multiwell Reader	Simultaneous read-out channels	**≥ 768 channels** acquired concurrently to support high-throughput, multi-well electrophysiological screening
5. Automated Electrophysiology Analytics	Pattern classification accuracy	**≥ 90% accuracy** in classification of drug-induce distinguishing drug- induced electrophysiological phenotypes based on validated reference datasets

R&D Plan	**Phase 1: Single-Well Feasibility Validation and Core Analytics Engine Development (Months 1–6)** • Conduct single-well 3D MEA feasibility studies using human cardiac and neuronal organoids • Acquire high-quality electrophysiological data under control and drug-treated conditions • Develop and prototype automated electrophysiology analysis pipelines for signal preprocessing and pattern classification • Benchmark performance and define quantitative specifications for multi-well system development • Conduct a phase-gate review to determine readiness for Phase 2 **Phase 2: Multi-Well System Development and Commercialization-Oriented Expansion (Months 7–38)** • Develop scalable 2D and 3D multi-well MEA plate designs • Design and implement a high-throughput MEA reader system meeting defined performance targets • Integrate GPU-accelerated signal processing architectures • Expand electrophysiology analytics to support multi-well datasets and 3D electrophysiological conduction mapping • Validate system performance using human organoid–based functional pharmacology assays • Complete system integration and achieve TRL 6

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The international research institute will focus on scientific feasibility validation and risk reduction, including: • Provision of research-grade single-well 3D MEA device concepts and design expertise • Establishment and operation of human cardiac and neuronal organoid models • Execution of proof-of-concept electrophysiology experiments under drug exposure • Generation of high-quality experimental datasets and biological insights • Participation in phase-gate evaluations and definition of quantitative benchmarks • Knowledge transfer to support transition from research-grade to commercial-grade system design
Expected R&R of Korean SME Partner	The Korean SME partner will serve as the lead industrial entity, including: • Overall system architecture definition and product-oriented development leadership • Commercial-grade design and development of 2D and 3D MEA plates • Development of a high-throughput MEA reader system meeting performance targets • Development and integration of automated electrophysiology analytics engines • Implementation of GPU-accelerated parallel signal processing • System integration, testing, validation, and TRL advancement • Project management, IP strategy, and commercialization planning

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** • Establishment of a next-generation high-throughput electrophysiology platform integrating advanced MEA hardware and automated data analytics • Advancement of 3D MEA technologies from research-grade concepts to commercial-grade systems • Demonstration of human organoid–based functional pharmacology assessment with improved sensitivity and reproducibility **Commercial and Industrial Impact** • Creation of a differentiated electrophysiology product portfolio, including MEA plates, reader systems, and integrated analysis software • Reduced reliance on imported electrophysiology instrumentation through domestic technology localization • Establishment of scalable business models based on instruments, consumables, and software licensing **Broader Impact** • Support for global regulatory trends toward human-relevant, non-animal testing methodologies • Strengthening of Korea's competitiveness in bioinstrumentation, next-generation life science tools, and advanced in vitro testing platforms • Enablement of future global collaborations, technology transfer, and export-oriented growth

6. References

(1) U.S. Food and Drug Administration (FDA). Roadmap to Reducing Animal Testing in Preclinical Safety Studies. Silver Spring, MD: U.S. Food and Drug Administration; 2025.

(2) Choi SJ, Liu Z, Yang F, Wang H, George D, Gracias DH, Kim DH. 3D Spatiotemporal Electrophysiology of Cardiac Organoids Using Shell Microelectrode Arrays. Advanced Materials. 2025 Oct 7:e06793. doi:10.1002/adma.202506